



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO PKS FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

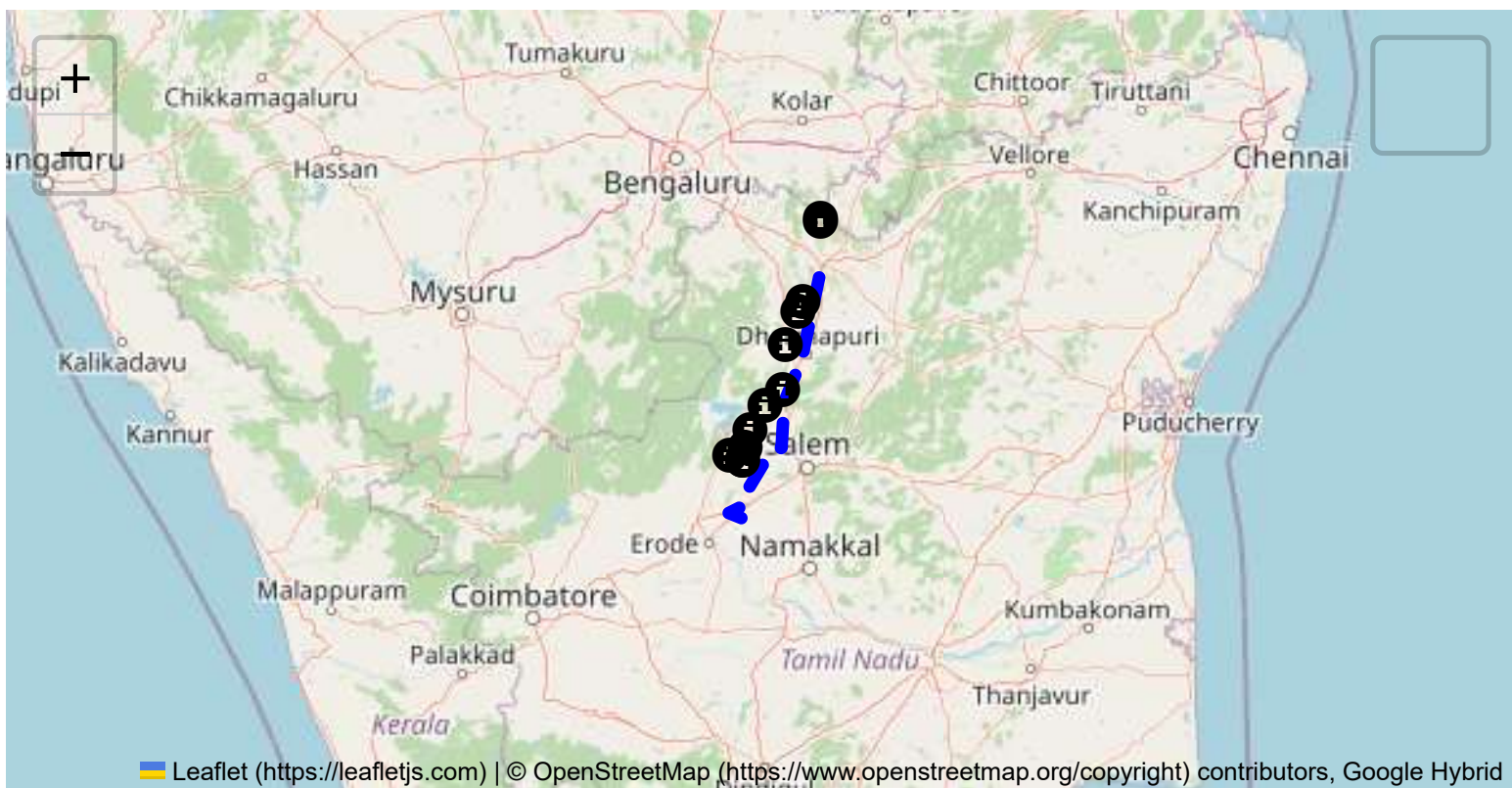
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

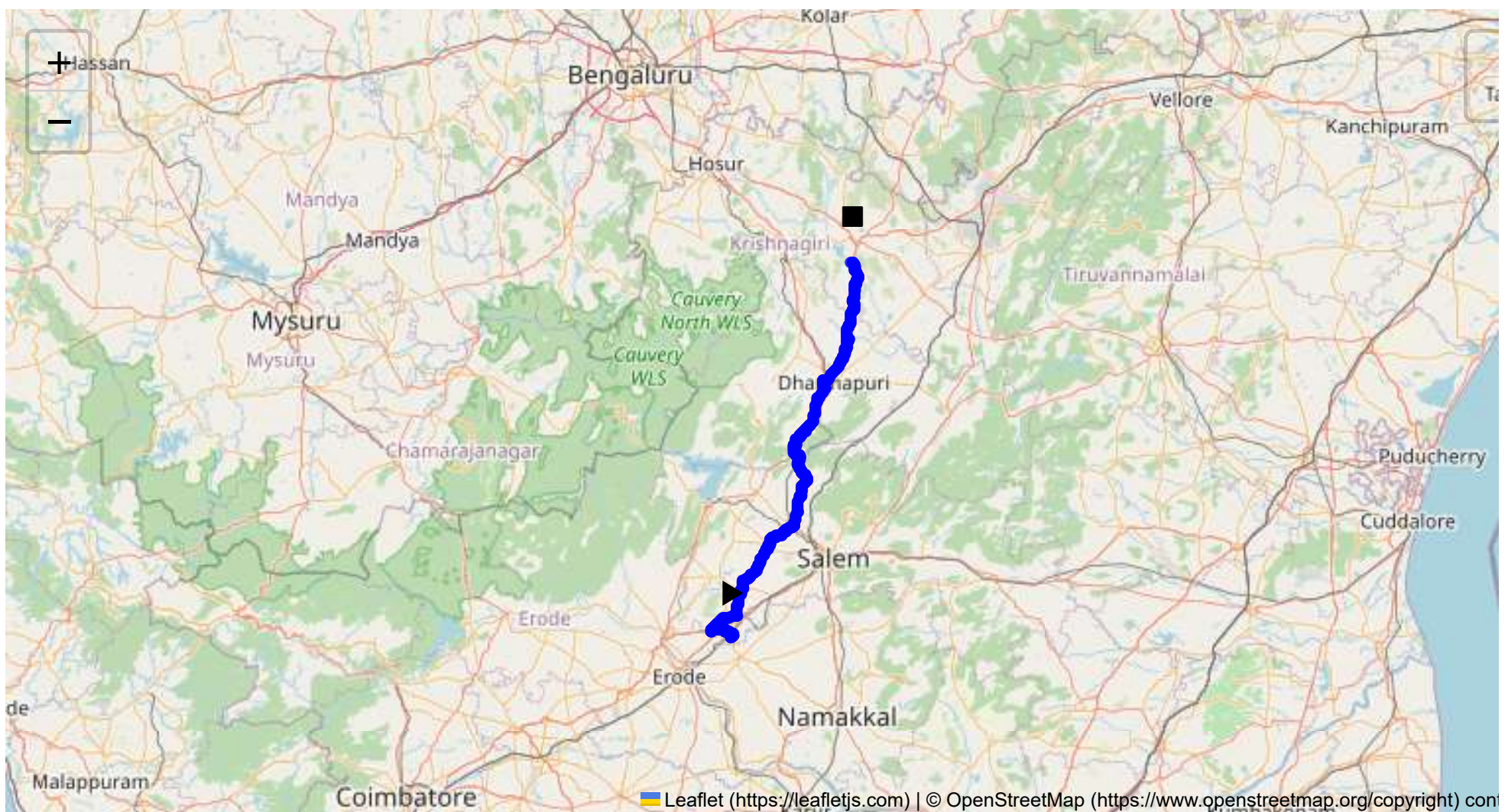
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 146.44 km
Estimated Duration: 2.9 hours
Adjusted Duration (Heavy Vehicle): 3.6 hours
Start: (11.4381, 77.8734)
End: (12.466239, 78.2065422)



Welcome to the Journey Risk Management Study

Route Overview: The route begins in Sangagiri, travels through Puthur, Padaiveedu, and other locales, ending in Periyamuthur near the KRP Dam. It spans about 146.44 km, primarily using a combination of national highways and state roads that connect several towns and rural areas.

Weather Conditions: Tamil Nadu experiences a tropical climate with distinct seasons. Between March and June, heatwaves can increase road surface temperatures, affecting vehicle performance. The monsoon

season (June to September) often brings heavy rains, potentially leading to flooding and reduced visibility. In winter (October to February), weather is milder, but early morning fog can be an issue.

Traffic Patterns:

- **Peak Hours:** Urban sections, particularly around Salem and Dharmapuri, witness high traffic volumes during traditional commuting hours (8-10 AM, 5-7 PM).
- **Congestion-Prone Areas:** Salem and stretches near major junctions on the Salem-Kochi Highway and Dharmapuri are congested during peak periods.

Road Quality and Infrastructure: Most highways on this route are in good condition, with well-maintained surfaces and clear signage. However, some rural stretches may have narrower, less maintained roads.

Alternative Routes for Emergencies: In case of road closures or other emergencies, diversions through regional roads connecting Salem to Dharmapuri, and then towards Hosur, can be considered. Precise navigation would depend on the actual point of emergency.

Local Regulations: Tamil Nadu has strict regulations on the transportation of hazardous materials, including mandatory permits, outlined routes, and times for heavy vehicle movement through certain urban areas to minimize risk to the public.

Historical Incidents: Heavy vehicle accidents are more common during monsoon seasons due to poor visibility and road conditions. Previous incidents have highlighted the need for caution around Salem and Dharmapuri, especially regarding sharp curves and slopes.

Environmental Considerations and Sensitive Areas: The area near KRP Dam and surrounding countryside in Tamil Nadu includes sensitive ecological zones. Drivers should minimize noise and air pollution around these areas to protect local wildlife and ecosystems.

Communication Coverage: Major stretches offer decent mobile coverage. However, certain rural areas between towns could have spotty network connections. Prior coordination with local networks could provide solutions for these gaps.

Emergency Response Times:

- **Sangagiri to Salem:** Approximately 20-30 minutes for emergency services due to proximity to urban centers.
- **Salem to Dharmapuri:** Around 45 minutes, slightly longer in remote or rural areas.
- **Dharmapuri to Periyamuthur:** Could take up to an hour due to limited infrastructure and distances from nearest emergency services.

Overall Risk Assessment Summary: The primary risks involve weather-related challenges, especially during the monsoon season, and potential congestion in urban areas. Road quality is generally reliable, but vigilance is necessary in areas with lesser road maintenance. Emergency response coverage is reasonable, though more remote areas could lead to delays. Adhering to local regulations and being prepared for sudden weather changes and spotty communication coverage remain essential for route safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43961, 77.87341	15 KM/Hr
1	Turn	High	11.44024, 77.87528	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	Medium	11.45543, 77.81740	30 KM/Hr
7	Turn	High	11.45584, 77.81695	15 KM/Hr
8	Turn	Medium	11.45656, 77.81709	30 KM/Hr
9	Turn	High	11.49383, 77.88442	15 KM/Hr
10	Turn	Medium	11.56083, 77.89921	30 KM/Hr
11	Turn	Medium	11.56180, 77.90019	30 KM/Hr
12	Turn	Medium	11.59189, 77.90645	30 KM/Hr
13	Turn	Medium	11.59328, 77.90809	30 KM/Hr
14	Turn	Medium	11.63867, 77.94841	30 KM/Hr
15	Turn	High	11.64217, 77.95200	15 KM/Hr
16	Turn	Medium	11.64712, 77.95118	30 KM/Hr
17	Turn	Medium	11.67858, 77.96834	30 KM/Hr
18	Turn	High	11.70233, 77.97869	15 KM/Hr
19	Turn	Medium	11.71394, 77.99572	30 KM/Hr
20	Turn	Medium	11.71916, 78.01117	30 KM/Hr
21	Turn	Medium	11.72504, 78.01680	30 KM/Hr
22	Turn	Medium	11.73766, 78.03445	30 KM/Hr
23	Turn	High	11.73651, 78.03868	15 KM/Hr
24	Turn	Medium	11.74257, 78.04708	30 KM/Hr
25	Turn	Medium	11.86805, 78.08353	30 KM/Hr
26	Turn	Medium	11.87913, 78.08206	30 KM/Hr
27	Turn	Medium	11.92566, 78.06503	30 KM/Hr
28	Turn	Medium	11.93694, 78.05425	30 KM/Hr
29	Turn	Medium	11.95531, 78.04698	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
30	Turn	High	11.95803, 78.04964	15 KM/Hr
31	Turn	Medium	11.96068, 78.04960	30 KM/Hr
32	Turn	High	11.96370, 78.05142	15 KM/Hr
33	Turn	High	11.96542, 78.04938	15 KM/Hr
34	Turn	High	11.96650, 78.04927	15 KM/Hr
35	Turn	Medium	12.12094, 78.13269	30 KM/Hr
36	Turn	High	12.14089, 78.12817	15 KM/Hr
37	Turn	High	12.14056, 78.13957	15 KM/Hr
38	Turn	Medium	12.24611, 78.19519	30 KM/Hr
39	Turn	Medium	12.25200, 78.20034	30 KM/Hr
40	Turn	Medium	12.40026, 78.21775	30 KM/Hr
41	Turn	Medium	12.40734, 78.22344	30 KM/Hr
42	Turn	High	12.46292, 78.21416	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
17	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
18	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
19	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
20	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
21	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
26	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
27	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
29	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
28	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
30	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44024, 77.87528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45764, 77.85219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45584, 77.81695



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45656, 77.81709



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49383, 77.88442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56083, 77.89921



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56180, 77.90019



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59189, 77.90645



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59328, 77.90809



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63867, 77.94841



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64217, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64712, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70233, 77.97869



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71394, 77.99572



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71916, 78.01117



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72504, 78.01680



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73766, 78.03445



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03868



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74257, 78.04708



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86805, 78.08353



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87913, 78.08206



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.92566, 78.06503



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93694, 78.05425



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95531, 78.04698



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95803, 78.04964



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96068, 78.04960



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96370, 78.05142



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96650, 78.04927



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14089, 78.12817



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.14056, 78.13957



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.24611, 78.19519



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.25200, 78.20034



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.40026, 78.21775



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.40734, 78.22344



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.46292, 78.21416
